

Coding & Solution

Date	03 October 2026
Team ID	4
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Coding & Solution

FitBuddy implements all four scenarios of the project brief and fixes the brief's known defects: manually typed user IDs, ignored age/weight, overwritten revisions, raw markdown output and error text saved as plans.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/sharukeshalagiri/FitBuddy
Programming Language(s)	Python 3.11+, HTML5, CSS3, Jinja2 templating
Framework(s) Used	FastAPI, Uvicorn, SQLAlchemy 2.x, Pydantic v2, google-genai SDK
Key Features Implemented	- Personalized 7-day plan (age, weight, goal, intensity, experience) as structured JSON → day cards - Nutrition / recovery tip (light model, built-in fallback tip) - Feedback revision of the latest version with full version history - Password-protected admin dashboard with original vs latest plan and delete - JSON API with Swagger docs; health endpoint - Retry + automatic fallback model; demo mode with no API key
Pending / Incomplete Features	None from the planned scope. Future scope: meal plans, progress tracking, cloud deployment (see Scalability & Future Plan).
Setup / Run Instructions	<pre>python -m venv venv venv\Scripts\activate (Mac/Linux: source venv/bin/activate) pip install -r requirements.txt copy .env.example .env (then add GEMINI_API_KEY; optional) uvicorn app.main:app --reload Open http://127.0.0.1:8000 API docs: http://127.0.0.1:8000/docs</pre>

Representative code (simplified from app/gemini_generator.py)

```
def generate_workout_gemini(user: UserInput) -> GeneratedPlan:
    """Generate a validated 7-day plan using the workout model (Pro role)."""
    prompt = build_workout_prompt(
        # uses age, weight, goal,
        age=user.age, weight_kg=user.weight, # intensity AND experience
        goal=user.goal, intensity=user.intensity,
        experience=user.experience)
    plan, model_used = call_structured(
        # retry, then fallback model
        model=settings.workout_model,
        fallback=settings.workout_fallback_model,
        prompt=prompt, schema=WorkoutPlan) # JSON response_schema
    validate_plan(plan) # exactly 7 days, >=1 rest day
    return GeneratedPlan(plan=plan, model=model_used)
```

Code Quality Checklist

S.No	Criteria	Status
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

- 35 automated tests (pytest) pass in demo mode, covering every HTML route, JSON endpoint, validation, admin auth, versioning and the fallback model.
- Real-Gemini run verified: schema accepted, every plan passed validation; revisions applied feedback (e.g. "no equipment" switched to bodyweight moves).
- The project brief's Gemini 1.5 models and google-generativeai SDK are retired/deprecated; FitBuddy uses current models and the google-genai SDK.
- Development was assisted by an AI coding assistant (Claude Code); the team specified, reviewed and tested the implementation.